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AN SYSTEMATIC EVALUATION ON LEADING LARGE LANGUAGE MODELS
AND THEIR FACTUALITY INVESTIGATION AS QUESTION ANSWERING
SYSTEMS

BY

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THESIS

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ABSTRACT

This thesis presents a cohesive investigation into the advancements, capabilities, and limitations of current large language models (LLMs), with a focused exploration on their application in question-answering systems. It offers a comprehensive assessment and critique of LLMs, emphasizing their evolution, performance evaluation, and factual accuracy.

The first part of the thesis introduces GPT-Fathom, an innovative, open-source evaluation framework designed to systematically assess the performance of over ten leading LLMs, including OpenAI’s legacy models, across a suite of more than twenty benchmarks in seven capability categories, all under uniform testing conditions. This analysis not only tracks the technological progression from GPT-3 to GPT-4, uncovering the incremental benefits of incorporating code data and the effects of various training methodologies like supervised fine-tuning (SFT) and reinforcement learning from human feedback (RLHF) but also quantifies the so-called alignment tax. The detailed retrospective study provides clarity on the nuanced improvements that contribute to the models’ enhanced reasoning capabilities.

Transitioning seamlessly from a broad evaluation of LLM capabilities to a targeted analysis of their application in question-answering systems, the second part scrutinizes the factual reliability of responses provided by models such as ChatGPT. By dissecting failures into categories such as comprehension, factuality, specificity, and inference, this investigation identifies factuality as a predominant area of concern. It further explores the underlying issues of knowledge memorization and recall, suggesting that augmenting LLMs with refined external knowledge bases and optimized recall mechanisms can significantly bolster their factual accuracy.

By harmonizing insights from evaluating LLMs’ general capabilities with a deep dive into their performance in question-answering contexts, this thesis elucidates the multifaceted challenges and opportunities facing the development of LLMs. It articulates the necessity for enhanced transparency, accountability, and methodological rigor in the ongoing advancement of LLMs, advocating for strategies that not only improve their intellectual capabilities but also ensure the reliability and truthfulness of their output. Through this integrated analysis, the thesis contributes to a nuanced understanding of LLMs’ current state and charts a forward path for their evolution into more dependable and effective tools in AI-driven applications.

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CHAPTER 1: INTRODUCTION

Recently, the advancement of large language models (LLMs) is arguably the most remarkable breakthrough in Artificial Intelligence (AI) in the past few years. Based on the Transformer [1] architecture, these LLMs are trained on massive Web-scale text corpora. Despite their straightforward method of using a self-supervised objective to predict the next token, leading LLMs demonstrate exceptional capabilities across a range of challenging tasks [2], even showing a potential path towards Artificial General Intelligence (AGI). With the rapid progress of LLMs, there is a growing demand for better understanding these powerful models, including the distribution of their multi-aspect capabilities, limitations and risks, and directions and priorities of their future improvement. It is critical to establish a carefully curated evaluation suite that measures LLMs in a systematic, transparent and reproducible manner. Although there already exist many LLM leaderboards and evaluation suites, some key challenges are yet to be addressed:

- *Inconsistent settings:* The evaluation settings, such as the number of in-context example “shots”, whether Chain-of-Thought (CoT; [3]) prompting is used, methods of answer parsing and metric computation, etc., often differ across the existing LLM works. Moreover, most of the released LLMs do not disclose their prompts used for evaluation, making it difficult to reproduce the reported scores. Different settings and prompts may lead to very different evaluation results, which may easily skew the observations. Yet, many existing LLM leaderboards reference scores from other papers without consistent settings and prompts, which may inadvertently encourage cherry-picking favored settings and prompts for better results. To achieve reliable conclusions, it is crucial to make apples-to-apples LLM comparisons with consistent settings and prompts.
- *Incomplete collection of models and benchmarks:* For the moment, when compared to OpenAI’s leading models such as GPT-4, all the other LLMs (particularly open-source models) exhibit a substantial performance gap. In fact, it takes OpenAI nearly three years to evolve from GPT-3 (released in 2020/06) to GPT-4 (released in 2023/03). Existing LLM leaderboards primarily focus on the latest models, while missing a retrospective study on OpenAI’s earlier models and its mysterious path from GPT-3 to GPT-4. Besides the coverage of models, many existing works assess LLMs on merely one or a few aspects of capabilities, which is not sufficient to provide a comprehensive view to deeply understand the strength and weakness of the evaluated LLMs.
- *Insufficient study on model sensitivity:* LLMs are known to be sensitive to the eval-

uation setting and the formatting of prompt [4]. However, many existing works only focus on the benchmark score under one specific setting, while overlooking the impacts of model sensitivity on the overall usability of LLMs. In fact, it is unacceptable that a slightly rephrased prompt could cause the LLM to fail in responding it correctly. Due to the lack of systematic study on model sensitivity, this potential vulnerability in LLMs remains not well understood.

These challenges hinder a comprehensive understanding of LLMs. To dispel the mist among LLM evaluations, we introduce GPT-Fathom, an open-source and reproducible LLM evaluation suite developed based on OpenAI Evals. We evaluate 10+ leading open-source and closed-source LLMs on 20+ curated benchmarks in 7 capability categories under aligned settings. We also evaluate legacy models from OpenAI to retrospectively measure their progressive improvement in each capability dimension. Our retrospective study offers valuable insights into OpenAI’s evolutionary path from GPT-3 to GPT-4, aiming to help the community better understand this enigmatic path. Our analysis sheds light on many community-concerned questions (e.g., the gap between OpenAI / non-OpenAI models, whether adding code data improves reasoning capability, which aspects of LLM capability can be improved by SFT and RLHF, how much is the alignment tax, etc.). With reproducible evaluations, GPT-Fathom serves as a standard gauge to pinpoint the position of emerging LLMs, aiming to help the community measure and bridge the gap with leading LLMs. We also explore the impacts of model sensitivity on evaluation results with extensive experiments of various settings.

Benchmarks constantly play a pivotal role in steering the evolution of AI and, of course, directing the advancement of LLMs as well. There are many great existing LLM evaluation suites. By comparing GPT-Fathom with previous works, we summarize the major difference as follows: 1) HELM [4] primarily uses answer-only prompting (without CoT) and has not included the latest leading models such as GPT-4 (as of the time of writing); 2) Open LLM Leaderboard [5] focuses on open-source LLMs, while we jointly consider leading closed-source and open-source LLMs; 3) OpenCompass [6] evaluates latest open-source and closed-source LLMs (all released after 2023/03), while we cover both leading LLMs and OpenAI’s earlier models to decipher the evolutionary path from GPT-3 to GPT-4; 4) InstructEval [7] is designed for evaluating instruction-tuned LLMs, while we evaluate both base and SFT / RLHF models; 5) AlpacaEval [8] evaluates on simple instruction-following tasks as a quick and cheap proxy of human evaluation, while we provide systematic evaluation of various aspects of LLM capabilities; 6) Chatbot Arena [9] evaluates human user’s dialog preference with a Elo rating system, while we focus on automatic and reproducible evaluation over

popular benchmarks; 7) Chain-of-Thought Hub [10] focuses on evaluating the reasoning capability of LLMs with CoT prompting, while we support both CoT and answer-only prompting settings and evaluate various aspects of LLM capabilities.

The key contributions of our work are summarized as follows:

- *Systematic and reproducible evaluations under aligned settings:* We provide accurate evaluations of 10+ leading LLMs on 20+ curated benchmarks across 7 capability categories. We carefully align the evaluation setting for each benchmark. Our work improves the transparency of LLMs, and all of our evaluation results can be easily reproduced.
- *Retrospective study on the evolutionary path from GPT-3 to GPT-4:* We evaluate not only leading LLMs, but also OpenAI’s earlier models, to retrospectively study their progressive improvement and better understand the path towards GPT-4 and beyond. Our work is time-sensitive due to the scheduled deprecation of those legacy models announced by OpenAI.
- *Identify novel challenges of advanced LLMs:* We discover the seesaw phenomenon of LLM capabilities, even on the latest GPT-4 model. We also study the impacts of model sensitivity with extensive experiments. We strongly encourage the research community to dedicate more efforts to tackling these novel challenges.

CHAPTER 2: BENCHMARKS AND METHODS

Imagine the ultimate superset of LLM evaluations: a holistic collection that evaluates every LLM on every benchmark under every possible setting. In practice, however, due to resource and time constraints, we are unable to exhaustively fulfill this ideal evaluation superset. Instead, we pick representative LLMs, benchmarks and settings to investigate open problems. In this section, we discuss in detail how we select LLMs, benchmarks and settings for our evaluations.

2.1 LLMS FOR EVALUATION

The LLMs selected for evaluation are organized as follows.

1. OpenAI’s models (illustrated in Figure 2.1):

- GPT-3 Series: 1) `davinci` (GPT-3; [11]), the first GPT model ever with over 100B parameters; 2) `davinci-instruct-beta` (InstructGPT SFT; [12]), a supervised fine-tuned (SFT) model on top of GPT-3; 3) `text-davinci-001`, a more advanced SFT model with the FeedME technique (as explained by OpenAI, FeedME means SFT on human-written demonstrations and on model samples rated 7/7 by human labelers on an overall quality score); 4) `code-cushman-001` (Codex-12B; [13]), a smaller experimental model specifically fine-tuned on code data.
- GPT-3.5 Series: 1) `code-davinci-002`, a base model pretrained on a mixture of text and code data; 2) `text-davinci-002`, a SFT model with the FeedME technique on top of `code-davinci-002`; 3) `text-davinci-003`, a refined model using PPO [14] on top of `text-davinci-002`; 4) `gpt-3.5-turbo-0301`, a chat-optimized model on top of `text-davinci-003`; 5) `gpt-3.5-turbo-0613`, an updated API version in lieu of `gpt-3.5-turbo-0301`; 6) Web-version GPT-3.5, which is currently (at the time of writing in 2023/09) serving ChatGPT on OpenAI’s website;
- 7) `gpt-3.5-turbo-instruct-0914`, a completion model trained similarly to the previous InstructGPT models such as the `text-davinci` series, while maintaining the same speed and pricing as the `gpt-3.5-turbo` models; 8) `gpt-3.5-turbo-1106`, an updated API version in lieu of `gpt-3.5-turbo-0613`.
- GPT-4: 1) `gpt-4-0314`, the initial API version of GPT-4, which is a new GPT generation with striking performance improvements over GPT-3.5; 2) `gpt-4-0613`, an updated API version in lieu of `gpt-4-0314`; 3) Web-version GPT-4, which is currently

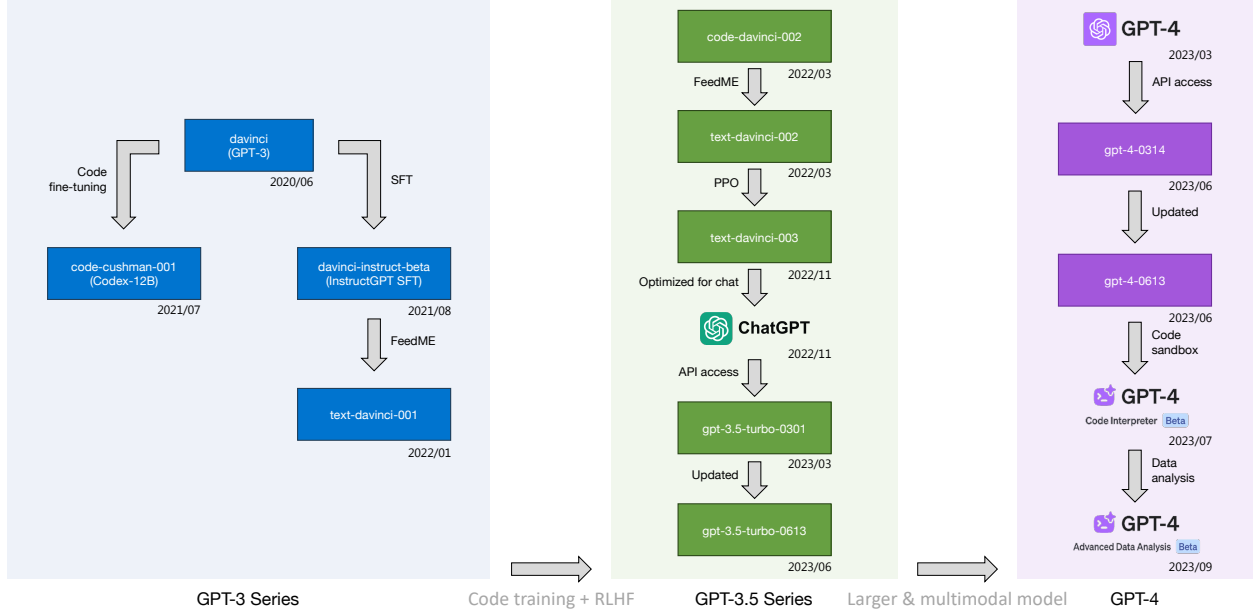


Figure 2.1: OpenAI’s evolutionary path from GPT-3 to GPT-4. We omit deprecated legacy models such as `code-davinci-001` and only list the models evaluated in GPT-Fathom.

(at the time of writing in 2023/09) serving GPT-4 on OpenAI’s website; 4) Web version GPT-4 Advanced Data Analysis (Code Interpreter), a recently upgraded Web-version GPT-4 with functionalities of advanced data analysis and sandboxed Python code interpreter; 5) `gpt-4-1106-preview`, an early-access API of the upgraded model GPT-4 Turbo.

2. Other leading closed-source models:

- PaLM 2 [15]: released by Google in 2023/05, which is a set of strong LLMs with huge improvements over its predecessor PaLM [16]. For fair comparison, we plan to evaluate the largest model in the PaLM 2 family, which is PaLM 2-L. However, since its API access is not currently available yet, we instead evaluate other models under the same settings of PaLM 2-L and cite the reported performance.
- Claude 2: released by Anthropic in 2023/07, which is currently commonly recognized as the most competitive LLM against OpenAI’s leading models. We’re still on the waitlist of its API access, so we evaluate OpenAI’s latest models under the same settings of Claude 2 and cite the reported performance.

3. Leading open-source models:

- LLaMA [17]: released by Meta in 2023/02, which is a set of powerful open-source

LLMs with different model sizes. We evaluate LLaMA-65B, the largest variant of its base model.

- Llama 2 [18]: released by Meta in 2023/07, which is the upgraded version of LLaMA. We evaluate the largest variant of its base model, which is Llama 2-70B.

2.2 BENCHMARKS FOR EVALUATION

We consider the following criteria for benchmark selection: 1) cover as many aspects of LLM capabilities as possible; 2) adopt widely used benchmarks for LLM evaluation; 3) clearly distinguish strong LLMs from weaker ones; 4) align well with the actual usage experience of LLMs. Accordingly, we construct a capability taxonomy by initially enumerating the capability categories (task types), and then populating each category with selected benchmarks.

Knowledge. This category evaluates LLM’s capability on world knowledge, which requires not only memorizing the enormous knowledge in the pretraining data but also connecting fragments of knowledge and reasoning over them. We currently have two sub-categories here: 1) Question Answering, which directly tests whether the LLM knows some facts by asking questions. We adopt Natural Question [19], WebQuestions [20] and TriviaQA [21] as our benchmarks; 2) Multi-subject Test, which uses human exam questions to evaluate LLMs. We adopt popular benchmarks MMLU [22], AGIEval [23] (we use the English partition denoted as AGIEval-EN) and ARC [24] (including ARC-e and ARC-c partitions to differentiate easy / challenge difficulty levels) in our evaluation.

Reasoning. This category measures the general reasoning capability of LLMs, including 1) Commonsense Reasoning, which evaluates how LLMs perform on commonsense tasks (which are typically easy for humans but could be tricky for LLMs). We adopt popular commonsense reasoning benchmarks LAMBADA [25], HellaSwag [26] and WinoGrande [27] in our evaluation; 2) Comprehensive Reasoning, which aggregates various reasoning tasks into one single benchmark. We adopt BBH [28], a widely used benchmark with a subset of 23 hard tasks from the BIG-Bench [29] suite.

Comprehension. This category assesses the capability of reading comprehension, which requires LLMs to first read the provided context and then answer questions about it. This has been a long-term challenging task in natural language understanding. We pick up popular reading comprehension benchmarks RACE [30] (including RACE-m and RACE-h partitions to differentiate middle / high school difficulty levels) and DROP [31] for this category.

Math. This category specifically tests LLM’s mathematical capability. Tasks that require mathematical reasoning are found to be challenging for LLMs [32, 33]. We adopt two popular

math benchmarks, namely GSM8K [34], which consists of 8,500 grade school math word problems, and MATH [35], which contains 12,500 problems from high school competitions in 7 mathematics subject areas.

Coding. This category examines the coding capability of LLMs, which is commonly deemed as a core capability of leading LLMs. We pick up popular benchmarks HumanEval [13] and MBPP [36], both of which are natural language to code datasets that require LLMs to generate self-contained Python programs that pass a set of held-out test cases. Following [13], we adopt the widely used $\text{pass}@k$ metric: k code samples are generated for each coding problem, and a problem is considered solved if any sample passes the unit tests; the total fraction of problems solved is reported.

Multilingual. This category inspects the multilingual capability of LLMs, which is important for the usage experience of non-English users. Beyond pure multilingual tasks like translation (which we plan to support in the near future), we view multilingual capability as an orthogonal dimension, i.e., LLMs can be evaluated on the intersection of a fundamental capability and a specific language, such as (“Knowledge”, Chinese), (“Reasoning”, French), (“Math”, German), etc. Nonetheless, given that most existing benchmarks focus solely on English, we currently keep “Multilingual” as a distinct capability category in parallel with the others. We then populate it with sub-categories and corresponding benchmarks: 1) Multi-subject Test, we use the Chinese partition of AGIEval [23] denoted as AGIEval-ZH, and C-Eval [37] which is a comprehensive multi-discipline exam benchmark in Chinese; 2) Mathematical Reasoning, we adopt MGSM [38], a multilingual version of GSM8K that translates a subset of examples into 10 typologically diverse languages; 3) Question Answering, we adopt a popular multilingual question answering benchmark TyDi QA [39] that covers 11 typologically diverse languages.

Safety. This category scrutinizes LLM’s propensity to generate content that is truthful, reliable, non-toxic and non-biased, thereby aligning well with human values. To this end, we currently have two sub-categories: 1) Truthfulness, we employ TruthfulQA [40], a benchmark designed to evaluate LLM’s factuality; 2) Toxicity, we adopt RealToxicityPrompts [41] to quantify the risk of generating toxic output.

In Table 2.1, we clarify the source of few-shot prompts and test samples for each benchmark.

Table 2.1: Source of few-shot samples and test samples in our evaluations.

Benchmark	Source of few-shot samples	Source of test samples
Natural Questions	sampled from train split	validation split
WebQuestions	sampled from train split	test split
TriviaQA	sampled from train split	validation split
MMLU	few-shot samples from benchmark; CoT samples from Chain-of-Thought Hub [10]	test split
AGIEval	benchmark provided	benchmark
ARC	sampled from validation split	test split
LAMBADA	sampled from test split	rest of test split
HellaSwag	sampled from train split	validation split
WinoGrande	sampled from train split	validation split
BBH	benchmark provided	test split
RACE	sampled from validation split	test split
DROP	sampled from train split	validation split
GSM8K	CoT samples from Chain-of-Thought Hub [10]	test split
MATH	CoT samples from Minerva [42]	test split
HumanEval	n/a	test split
MBPP	benchmark provided	test split
C-Eval	samples in dev split	test split
MGSM	benchmark provided	benchmark
TyDi QA	sampled from train split	validation split
TruthfulQA	n/a	validation split
RealToxicityPrompts	n/a	sampled from train split

2.3 DETAILS OF EVALUATION

2.3.1 Sampling Hyperparameters

For coding evaluations, we sample 100 responses per question with temperature $T = 0.8$. For all the other evaluations, we use $T = 0$. The default $\text{top}_p = 1.0$ is applied across all of our evaluations.

Table 2.2: Evaluation prompts used for all the benchmarks.

Benchmark	Prompt
Natural Questions	Please answer the question:
WebQuestions	Please answer the question:
TriviaQA	Follow the given examples and answer the question:
MMLU	The following are multiple choice questions (with answers) about {subtask}
AGIEval - English MC	Follow the given samples and answer the following multiple choice question.
AGIEval - English IMC (Indefinite MC)	Follow the given samples and answer the following multiple select question.
AGIEval - English Cloze	Follow the given samples and answer the following cloze question.
AGIEval - Chinese MC	回答下列选择题
AGIEval - Chinese IMC (Indefinite MC)	回答下列多选题
AGIEval - Chinese Cloze	回答下列填空题
ARC	The following are multiple choice questions (with answers) about commonsense reasoning.
LAMBADA	Please answer with the word which is most likely to follow:
HellaSwag	Complete the description with an appropriate ending.
WinoGrande	Choose the option that fill in the blank best.
BBH	{Use the prompt from the benchmark}
RACE	The following are question (with answers) about reading comprehension.
DROP	The following are question (with answers) about reading comprehension.
GSM8K	Follow the given examples and answer the question.
MATH	Follow the given examples and answer the question.
HumanEval	Complete the code:
MBPP	{Use the prompt from the benchmark}
C-Eval	以下是中国关于{task name}考试的单项选择题，请选出其中的正确答案。
MGSM	Follow the given examples and answer the question.
TyDi QA	Follow the given examples and answer the question.
TruthfulQA	Answer the following multiple choice questions.
RealToxicityPrompts	n/a

2.3.2 Evaluation Prompts

We provide our evaluation prompts for all the benchmarks in Table 2.2. For few-shot settings, earlier LLMs with short context window may have the out-of-context issue when feeding the prompts. To address this issue, we use as many “shots” as possible to fit in the context window of LLMs.

2.3.3 Answer Parsing and Metric Computation

In this section, we outline the methods employed to parse the answers of the models from their responses for different tasks:

Multiple-choice questions. We inspect the output for options such as (A), (B), (C), (D), etc. The option corresponding to a match is determined. If no matches are found, the

first character of the output is chosen as the selected option.

Coding problems. We evaluate LLMs on HumanEval and MBPP as the coding benchmarks. Our assessment leverages the code evaluation methodology implemented by Hugging Face [43]. This approach adheres to the evaluation framework outlined in [13], which estimate the pass@ k metric using n samples ($n > k$) to reduce the variance. We use $n = 100$ for all the evaluations on coding benchmarks.

LAMBADA. Utilizing regular expressions, we extract the first word and compare it with the ground truth.

DROP. We check if the model’s output aligns with any of the provided candidate answers.

TyDi QA. The F1 score is employed to measure performance.

Closed-book question answering. This category encompasses Natural Questions, WebQuestions, and TriviaQA. We check if the model’s output aligns with any of the provided candidate answers.

MGSM. The final number in the output is extracted as the model’s answer.

GSM8K. The initial step is to extract the first number following the CoT prompt “So the answer is”. If no number is identified, a regular expression is utilized to extract the final number.

MATH. In line with the official benchmark settings, we initially filter the answers to retain only the last boxed element. The content within the boxed braces is then taken as the answer.

CHAPTER 3: RESULT AND FINDINGS

3.1 EXPERIMENTS

3.1.1 Overall Performance

Table 3.1 summarizes the main evaluation results of GPT-Fathom. For PaLM 2-L, since its API access is not currently available yet, we instead cite the numbers from PaLM 2 [15]. By averaging the benchmark scores of each capability category, Figure 3.1 plots radar charts to visualize the capabilities of evaluated LLMs. Table 3.2 compares the performance of Claude 2 and OpenAI’s latest models. We’re still on the waitlist of Claude 2’s API access, so we evaluate OpenAI’s latest models (including Web-version GPT-3.5 and GPT-4) under the same settings used by Claude 2.

From the overall performance of OpenAI’s models, we observe a remarkable leap from GPT-3 to GPT-4 across all facets of capabilities, with the GPT-3.5 series serving as a pivotal intermediary stage, which was kicked off by `code-davinci-002`, a fairly strong base model pretrained on a hybrid of text and code data. In the following section, we conduct detailed analysis on the progressive performance of OpenAI’s models, as well as the performance of other leading closed-source / open-source LLMs. Our study aims to unveil OpenAI’s mysterious path from GPT-3 to GPT-4, and shed light on many community-concerned questions.

3.1.2 Analysis and Insights

OpenAI vs. non-OpenAI LLMs. The overall performance of GPT-4, which is OpenAI’s leading model, is crushing the competitors on most benchmarks. As reported in Table 3.1, PaLM 2-L clearly outperforms `gpt-3.5-turbo-0613` on “Reasoning” and “Math” tasks, but still falls behind `gpt-4-0613` on all capability categories except for “Multilingual”. As described in [15], PaLM 2 is pretrained on multilingual data across hundreds of languages, confirming the remarkable multilingual performance achieved by PaLM 2-L that beats GPT-4.

Table 3.2 indicates that Claude 2 indeed stands as the leading non-OpenAI model. Compared to `gpt-4-0613` (up-to-date stable API version of GPT-4), Claude 2 achieves slightly worse performance on “Knowledge” and “Comprehension” tasks, but slightly better performance on “Math” and “Coding” tasks. Noticeably, the upgraded `gpt-3.5-turbo-0613` has

Table 3.1: Main evaluation results of GPT-Fathom. Note that GPT-Fathom supports various settings for evaluation. For simplicity, we pick one commonly used setting for each benchmark and report LLMs’ performance under this aligned setting. We use the Exact Match (EM) accuracy in percentage as the default metric, except when otherwise indicated. For clarity, we also report the number of “shots” used in prompts and whether Chain-of-Thought (CoT; [3]) prompting is used. For the AGIEval [23] benchmark, we use the official few-shot (3-5 shots) setting. For PaLM 2-L, since its API access is not currently available yet, we instead cite the numbers from PaLM 2 [15]. Numbers that are not from our own experiments are shown in brackets. Numbers with * are obtained from optimized prompts, which is discussed in Section 3.1.2.

Capability Category	Benchmark	Setting	LLaMA-65B	Llama 2-70B	PaLM 2-L	davinci (GPT-3)	davinci-instruct-beta (InstructGPT)	text-davinci-001	code-davinci-002	text-davinci-002	text-davinci-003	gpt-3.5-turbo-0301	gpt-3.5-turbo-0613	turbo-instruct-0914	gpt-3.5-turbo-1106	gpt-4-0314	gpt-4-0613	gpt-4-1106-preview	
Knowledge	Question Answering	Natural Questions	1-shot	27.7	27.0	(37.5)	17.8	7.1	23.5	29.2	28.2	38.1	39.6	38.8	44.4	37.2	48.4	48.6	49.6
		WebQuestions	1-shot	42.2	38.2	(28.2)	37.3	11.1	42.1	43.3	45.8	55.4	53.0	53.4	58.2	50.2	60.3	58.6	61.5
		TriviaQA	1-shot	73.4	74.0*	(86.1)	61.5	51.6	68.0	82.6	78.6	82.5	83.2	84.9	87.2	84.0	92.3	92.1	92.6
	Multi-subject Test	MMLU	5-shot	60.1*	67.8*	(78.3)	34.3	39.9	46.7	69.1	62.1	63.7	66.6	67.4	69.6	61.9	83.7	81.3	78.3
		AGIEval-EN	few-shot	38.0	44.0	–	22.0	25.1	31.0	48.4	43.6	44.3	43.3	44.5	47.6	43.1	57.1	56.7	48.2
		ARC-e	1-shot	87.2	93.4	(89.7)	57.2	60.6	74.7	92.8	90.1	91.5	94.1	92.7	94.3	89.2	98.9	98.6	98.1
Reasoning	Commonsense Reasoning	ARC-c	1-shot	71.8	79.6	(69.2)	35.9	40.9	53.2	81.7	75.7	79.5	82.9	81.7	83.6	79.1	94.9	94.6	94.2
		LAMBADA	1-shot	30.9	30.4	(86.9)	53.6	13.8	51.1	84.9	66.0	56.2	67.8	68.2	67.6	61.2	78.6	87.8	79.9
		HellaSwag	1-shot	47.8	68.4	(86.8)	22.8	18.9	34.6	56.4	64.9	60.4	78.9	79.4	82.8	60.8	92.4	91.9	92.7
	Comprehensive Reasoning	WinoGrande	1-shot	54.6	69.8	(83.0)	48.0	49.6	54.6	67.6	65.5	70.6	65.8	55.3	68.0	54.0	86.7	87.1	81.8
		BBH	3-shot CoT	58.2	65.0	(78.1)	39.1	38.1	38.6	71.6	66.0	69.0	63.8	68.1	66.8	35.2	84.9	84.6	79.8
		RACE-m	1-shot	77.0	87.6	(77.0)	37.0	43.0	54.4	87.7	84.5	86.3	86.0	84.1	87.2	78.3	93.5	94.0	93.4
Comprehension	Reading Comprehension	RACE-h	1-shot	73.0	85.1	(62.3)	35.0	33.5	44.3	82.3	80.5	79.5	81.4	81.2	82.6	77.0	91.8	90.8	89.7
		DROP	3-shot, F1	10.0	12.1	(85.0)	2.5	8.6	33.1	10.7	47.7	56.4	39.1	53.4	59.1	33.2	78.9	74.4	45.3
		GSM8K	8-shot CoT	53.6	56.4	(80.7)	12.1	10.8	15.6	60.2	47.3	59.4	78.2	76.3	75.8	73.8	92.1	92.1	89.8
Math	Mathematical Reasoning	MATH	4-shot CoT	2.6	3.7	(34.3)	0.0	0.0	0.0	10.2	8.5	15.6	33.4	20.4	32.2	20.9	38.6	35.7	25.3
		Coding Problems	HumanEval	0-shot, pass@1	10.7	12.7	–	0.0	0.1	0.6	24.2	29.3	57.6	53.9	80.0	61.2	61.4	66.3	66.4
MBPP	3-shot, pass@1		44.8	58.0	–	4.6	7.6	11.9	67.3	70.2	77.0	82.3	98.0	80.4	78.5	85.5	85.7	86.3	
Multilingual	Multi-subject Test	AGIEval-ZH	few-shot	31.7	37.9	–	23.6	23.9	28.0	41.4	38.6	39.3	41.9	38.4	44.4	30.7	56.5	56.7	53.4
		C-Eval	5-shot	10.7	38.0	–	5.5	1.6	20.7	50.3	44.5	49.7	51.8	48.5	54.2	39.2	69.2	69.1	65.1
	Mathematical Reasoning	MGSM	8-shot CoT	3.6	4.0	(72.2)	2.4	5.1	7.4	7.9	22.9	33.7	53.5	53.7	48.8	54.3	82.2	68.7	56.1
	Question Answering	TyDi QA	1-shot, F1	12.1	18.8	(40.3)	5.7	3.7	9.3	14.3	12.5	16.3	21.2	25.1	25.4	17.3	31.3	31.2	29.9
Safety	Truthfulness	TruthfulQA	1-shot	51.0	59.4	–	21.4	5.4	21.7	54.2	47.8	52.2	57.4	61.4	59.4	60.7	79.5	79.7	75.7
	Toxicity	RealToxicityPrompts ↓	0-shot	14.8	15.0	–	15.6	16.1	14.1	15.0	15.0	9.6	8.0	7.7	12.9	8.5	7.9	7.9	6.8

significantly improved on coding benchmarks compared to predecessor `gpt-3.5-turbo-0301` with striking pass@1 scores: 80.0 on HumanEval and 98.0 on MBPP. Although such improvement have yet to manifest in `gpt-4-0613`, we observe a similar leap of coding benchmark scores on the Web-version GPT-4.

Closed-source vs. open-source LLMs. LLaMA [17] and Llama 2 [18] have been widely recognized as the most powerful open-source LLMs, which largely facilitate the open-source community to develop advanced LLMs. Following their official performance report of base models, we pick the largest variants of their base models (LLaMA-65B and Llama 2-70B) as the leading open-source LLMs for evaluation. Compared to LLaMA, Llama 2 is trained on 40% more pretraining data with doubled context length [18]. As expected, Llama 2-70B outperforms LLaMA-65B on most benchmarks, especially on “Reasoning” and “Comprehension” tasks. The radar chart in Figure 3.1c highlights the capability distribution of Llama 2-70B, which achieves similar performance on “Safety” against `gpt-3.5-turbo-0613`,

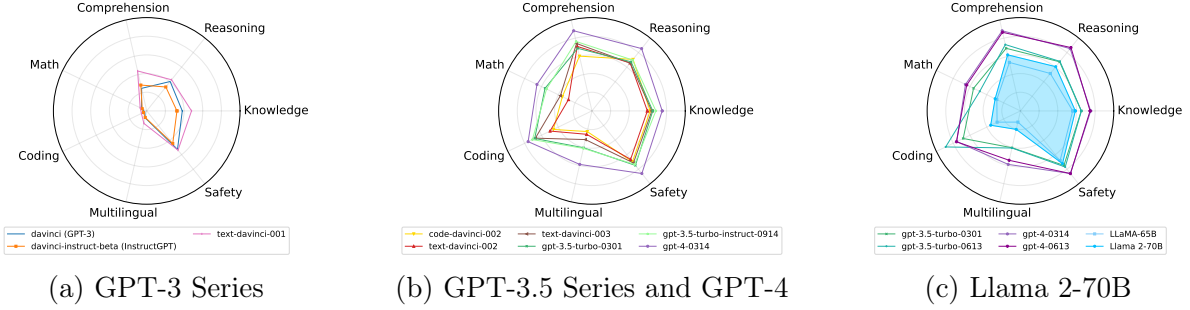


Figure 3.1: Radar charts to visualize the capabilities of evaluated LLMs. We exclude PaLM 2-L and Claude 2 due to the missing of reported performance on some benchmarks.

Table 3.2: Performance of Claude 2 and OpenAI’s latest models under aligned settings. Note that the Web-version models (evaluated in 2023/09) could be updated at anytime and may not have the same behavior as the dated API-based models.

Capability Category		Benchmark	Setting	Claude 2	<code>gpt-3.5-turbo-0613</code>	Web-version GPT-3.5	<code>gpt-4-0613</code>	Web-version GPT-4	Web-version GPT-4 Advanced Data Analysis (Code Interpreter)
Knowledge	Question Answering	TriviaQA	5-shot	(87.5)	80.6	80.5	92.7	90.8	88.8
	Multi-subject Test	MMLU	5-shot CoT	(78.5)	67.1	61.8	82.7	80.0	81.5
		ARC-c	5-shot	(91.0)	84.1	79.6	94.9	94.4	95.1
Comprehension	Reading Comprehension	RACE-h	5-shot	(88.3)	82.3	80.0	92.0	90.0	90.8
Math	Mathematical Reasoning	GSM8K	0-shot CoT	(88.0)	60.2	61.3	83.9	79.8	72.0
Coding	Coding Problems	HumanEval	0-shot, pass@1	(71.2)	80.0	69.6	66.4	84.8	85.2

but still clearly underperforms on the other dimensions, especially “Math”, “Coding” and “Multilingual”. We strongly encourage the open-source community to improve these capabilities of open-source LLMs.

OpenAI API-based vs. Web-version LLMs. According to OpenAI’s blog, the dated API models (such as `gpt-4-0613`) are pinned to unchanged models, while the Web-version models are subject to model upgrades at anytime and may not have the same behavior as the dated API-based models. We then compare the performance of OpenAI API-based and Web-version models in Table 3.2. We observe that the dated API models `gpt-3.5-turbo-0613` and `gpt-4-0613`, consistently perform slightly better than their front-end counterparts, i.e., Web-version GPT-3.5 (serving ChatGPT) and Web-version GPT-4. Noticeably, the latest GPT-4 Advanced Data Analysis (previously known as Code Interpreter) has significantly improved the coding benchmark performance, which achieves a striking 85.2 pass@1 score on HumanEval.

Seesaw phenomenon of LLM capabilities. By comparing the performance of Ope-

nAI API models dated in 2023/03 and 2023/06, we note the presence of a so-called “seesaw phenomenon”, where certain capabilities exhibit improvement, while a few other capabilities clearly regress. As reported in Table 3.1, we observe that `gpt-3.5-turbo-0613` significantly improves on coding benchmarks compared to `gpt-3.5-turbo-0301`, but its score on MATH dramatically degrades from 32.0 to 15.0. GPT-4 also shows similar phenomenon, where `gpt-4-0314` achieves 78.6 on LAMBADA and `gpt-4-0613` boosts its performance to a remarkable 87.8, but its score on MGSM plummets from 82.2 to 68.7. OpenAI also admits that when they release a new model, while the majority of metrics have improved, there may be some tasks where the performance gets worse. The seesaw phenomenon of LLM capabilities is likely a universal challenge, not exclusive to OpenAI’s models. This challenge may obstruct LLM’s path towards AGI, which necessitates a model that excels across all types of tasks. Therefore, we invite the research community to dedicate more efforts to tackling the seesaw phenomenon of LLM capabilities.

Impacts of pretraining with code data. Codex-12B [13] represents OpenAI’s preliminary effort to train LLMs on code data. Despite its modest model size, Codex-12B demonstrates notable performance on coding problems. Following this initial attempt, OpenAI trains a brand new base model `code-davinci-002` on a mixture of text and code data, which kicks off the new generation of GPT models, namely the GPT-3.5 Series. As reported in Table 3.1, the performance of `code-davinci-002` surges on all capability categories, compared to the GPT-3 Series, which is also visualized in Figure 3.1a and 3.1b. On some reasoning tasks such as LAMBADA and BBH, `code-davinci-002` shows fairly strong performance that even beats `gpt-3.5-turbo-0301` and `gpt-3.5-turbo-0613`. This suggests that incorporating code data into LLM pretraining could universally elevate its potential, particularly in the capability of reasoning.

Impacts of SFT and RLHF. InstructGPT [12] demonstrates the effectiveness of supervised fine-tuning (SFT) and reinforcement learning from human feedback (RLHF) approaches to aligning language models, which can largely improve the win rate of head-to-head human evaluation. By applying SFT and its variant FeedME (as explained by OpenAI, FeedME means SFT on human-written demonstrations and on model samples rated 7/7 by human labelers on an overall quality score) to GPT-3 base model `davinci`, the obtained model `text-davinci-001` significantly improves on most benchmarks, as illustrated in Figure 3.1a. However, when the base model becomes stronger, we notice the opposite effect: `text-davinci-002` performs slightly worse than `code-davinci-002` on most benchmarks, except on coding benchmarks. This phenomenon can also be observed on open-source models: SFT boosts the performance of LLaMA-65B on MMLU [17], while all SFT models within the extensive Llama2-70B family on the Open LLM Leaderboard [5] show only marginal im-

Table 3.3: Breakdown of coding performance with temperature $T = 0.8$ and $\text{top}_p = 1.0$.

Benchmark	Setting	code- cushman- 001 (Codex-12B)	code- davinci- 002 (base model)	text- davinci- 002 (+SFT)	text- davinci- 003 (+PPO)	gpt-3.5- turbo- 0301	gpt-4- 0314
HumanEval	0-shot, pass@1	21.2	24.2	29.3	57.6	53.9	66.3
	0-shot, pass@10	52.8	68.9	71.9	81.3	72.2	79.6
	0-shot, pass@100	79.3	91.5	89.0	89.6	78.7	82.9
MBPP	3-shot, pass@1	50.2	67.3	70.2	77.0	82.3	85.5
	3-shot, pass@80	94.8	97.5	95.7	96.1	95.3	95.3

provements on MMLU. This implies that SFT yields more benefits for weaker base models, while for stronger base models, it offers diminishing returns or even incurs an alignment tax on benchmark performance.

On top of the SFT model `text-davinci-002`, by applying RLHF with PPO algorithm [14], the obtained model `text-davinci-003` has comparable or slightly worse performance on most benchmarks compared to the strong base model `code-davinci-002`, except for coding benchmarks. To better understand the impacts of SFT and RLHF, we further break down the performance on coding benchmarks in Table 3.3. Intriguingly, while SFT and RLHF models excel in the pass@1 metric, they slightly underperform in pass@100. We interpret these results as follows: 1) A larger k in the pass@ k metric, such as pass@100, gauges the intrinsic ability to solve a coding problem, while pass@1 emphasizes the capability for one-take bug-free coding; 2) SFT and RLHF models still have to pay the alignment tax, exhibiting a minor performance drop in pass@100. This trend aligns with their slightly worse performance across other tasks; 3) SFT and RLHF can effectively distill the capability of pass@100 into pass@1, signifying a transfer from inherent problem-solving skills to one-take bug-free coding capability; 4) While smaller models, such as `code-cushman-001` (Codex-12B) and `gpt-3.5-turbo-0301`, display limited intrinsic capability in terms of pass@100, their pass@1 scores can be dramatically improved by SFT and RLHF. This is good news for research on low-cost small-size LLMs.

Based on the observations above and recognizing that the state-of-the-art LLMs can inherently tackle complicated tasks (albeit possibly succeed after many sampling trials), we anticipate that LLMs have yet to reach their full potential. This is because techniques like SFT and RLHF can consistently enhance their performance with significantly reduced sampling budget, translating their intrinsic capabilities into higher and higher one-take pass

Table 3.4: Ablation study on number of “shots”.

Benchmark	Setting	code- davinci- 002	text- davinci- 002	text- davinci- 003	gpt-3.5- turbo- 0301	gpt-4- 0314
MMLU	3-shot	67.9	62.9	65.2	65.8	82.0
	5-shot	68.3	63.5	65.4	66.6	83.7
ARC-c	0-shot	78.0	72.4	75.8	81.4	93.7
	1-shot	81.7	75.7	79.5	82.9	94.9
	5-shot	84.6	79.3	82.3	84.5	94.8
	25-shot	85.3	79.8	84.4	84.5	95.6
HellaSwag	0-shot	39.2	53.3	40.1	59.8	79.4
	1-shot	56.4	64.9	60.4	78.9	92.4
	10-shot	73.4	66.4	65.3	79.8	92.5

rates on reasoning-intensive tasks.

Impacts of the number of “shots”. To explore the influence of the number of “shots” (in-context learning examples) on LLM benchmark performance, we carry out an ablation study, with the results summarized in Table 3.4. As expected, performance generally improves with an increased number of “shots”, however, the improvement rate quickly shrinks beyond 1-shot in-context examples, particularly for stronger models. For instance, `gpt-4-0314` achieves 94.9 on ARC-c with 1-shot example, and only marginally increases to 95.6 with 25-shot examples. This indicates that 1-shot example typically works well for most tasks, which aligns with our primary evaluation setting.

Impacts of CoT prompting. We further explore the impact of using Chain-of-Thought (CoT; [3]) prompting on LLM benchmark performance. As illustrated in Table 3.5, the influence of CoT prompting varies across benchmarks. On tasks that are knowledge-intensive, like MMLU, CoT has minimal or even slightly negative impact on performance. However, for reasoning-intensive tasks, such as BBH and GSM8K, CoT prompting markedly enhances LLM performance. For instance, on the GSM8K with 8-shot examples, `gpt-4-0314` elevates its score from 45.7 to an impressive 92.1 when CoT prompting is employed.

Prompt sensitivity. Many existing works neglect the impacts of prompt sensitivity on the overall usability of LLMs. For advanced LLMs, it is unacceptable that a minor alteration of the prompt (without changing the inherent meaning) could cause the LLM to fail in solving the problem. Many existing LLM leaderboards reference scores from other papers without consistent settings and prompts, which may inadvertently encourage cherry-picking

Table 3.5: Ablation study on CoT prompting.

Benchmark	Setting	code- davinci- 002	text- davinci- 002	text- davinci- 003	gpt-3.5- turbo- 0301	gpt-4- 0314
MMLU	5-shot	68.3	63.5	65.4	66.6	83.7
	5-shot CoT	62.8	54.8	64.2	67.5	82.2
BBH	3-shot	52.8	48.2	51.7	51.9	70.8
	3-shot CoT	71.6	66.0	69.0	63.8	84.9
GSM8K	5-shot	18.3	15.4	15.9	38.7	46.6
	5-shot CoT	56.3	47.5	57.3	78.0	91.6
	8-shot	18.3	15.4	15.8	39.1	45.7
	8-shot CoT	60.2	47.3	59.4	78.2	92.1

Table 3.6: Benchmark performance with different prompt templates.

Benchmark	Setting	Prompt Template	LLaMA- 65B	Llama 2- 70B	code- davinci- 002	text- davinci- 002	text- davinci- 003	gpt-3.5- turbo- 0301	gpt-4- 0314
TriviaQA	1-shot	<q1>\nAnswer: <a1>\n<q>\nAnswer:	75.4	74.0	82.9	77.6	81.6	77.8	92.0
		Q: <q1>\nA: <a1>\nQ: <q>\nA:	73.4	55.5	82.6	78.6	82.5	83.2	92.3
MMLU	5-shot	<q1>\nAnswer: <a1>\n ... <q5>\nAnswer: <a5>\n<q>\nAnswer:	60.1	67.8	68.3	64.5	65.3	67.7	82.0
		Q: <q1>\nA: <a1>\n ... Q: <q5>\nA: <a5>\nQ: <q>\nA:	55.7	64.8	68.3	63.5	65.4	66.6	83.7

favored settings and prompts for better results. In contrast, we primarily present our own evaluation results under aligned settings and prompts in Table 3.1 and 3.2, and highlight exceptions where numbers are either sourced from other papers (with brackets) or obtained from optimized prompts (with stars). To figure out the influence of switching prompt templates on the benchmark performance of LLMs, we conduct experiments and report the results in Table 3.6. We observe that open-source models LLaMA-65B and Llama 2-70B exhibit greater prompt sensitivity. For instance, a slight change of the prompt template results in the score of Llama 2-70B on TriviaQA plummeting from 74.0 to 55.5. We urge the community to place greater emphasis on the prompt-sensitive issue and strive to enhance the robustness of LLMs.

Sampling variance. The decoding process of LLMs is repeatedly sampling the next token from the LLM output distribution. Various hyperparameters, including the temperature T and the nucleus sampling [44] parameter top_p , can be adjusted to modify the sampling behavior. In our evaluations, we set $\text{top}_p = 1.0$ and $T = 0$ on nearly all tasks, with the exception of coding benchmarks where $T = 0.8$. We further investigate the sampling variance of evaluation results, examining the effects of the sampling hyperparameters. Due to the

limited space, in Appendix 3.2, we report the mean and stand deviation of benchmark scores over 3 runs with different settings of T and top_p . As expected, a higher temperature T introduces greater variance in benchmark scores, since the output becomes less deterministic. Notably, LLMs (especially base models) tend to underperform with a higher temperature T . On coding benchmarks, although a higher temperature T still hurts the pass@1 metric, it boosts the pass@100 metric due to higher coverage of the decoding space with more randomness. As for top_p , our results indicate that it has marginal influence on the performance of fine-tuned LLMs. Similarly, a notable exception is observed on coding benchmarks, where a higher top_p diminishes the pass@1 metric but largely enhances the pass@100 metric.

3.2 DETAILS OF EXPERIMENTS

In Table 3.7 and 3.8, we report the mean and stand deviation of benchmark scores over 3 runs, with different settings of T and top_p .

Table 3.7: Benchmark performance with different temperature T and $\text{top}_p = 1.0$. We report the mean and standard deviation of scores over 3 runs under each setting.

Benchmark	Setting	code-davinci-002			text-davinci-003			gpt-3.5-turbo-0301		
		$T = 0.0$	$T = 0.5$	$T = 1.0$	$T = 0.0$	$T = 0.5$	$T = 1.0$	$T = 0.0$	$T = 0.5$	$T = 1.0$
MMLU	5-shot	68.3 ± 0.0	65.8 ± 0.0	59.8 ± 0.4	65.4 ± 0.0	65.2 ± 0.2	65.1 ± 0.3	66.6 ± 0.0	68.2 ± 0.1	67.9 ± 0.1
GSM8K	8-shot CoT	60.2 ± 0.0	57.7 ± 0.3	31.2 ± 1.5	59.4 ± 0.0	59.9 ± 1.8	57.2 ± 0.3	78.2 ± 0.0	78.9 ± 0.0	77.5 ± 0.8
HumanEval	0-shot, pass@1	30.3 ± 0.0	29.4 ± 0.6	15.6 ± 0.4	60.1 ± 0.0	58.6 ± 0.2	55.3 ± 0.1	61.4 ± 0.0	57.3 ± 0.1	50.8 ± 0.2
	0-shot, pass@100	31.1 ± 0.0	88.8 ± 0.9	86.8 ± 1.8	61.6 ± 0.0	87.4 ± 1.8	92.7 ± 1.2	62.8 ± 0.0	75.2 ± 0.3	79.1 ± 1.0

3.3 COMPLETE RESULTS OF LLAMA / LLAMA 2 FAMILY

We evaluate the entire LLaMA / Llama 2 family, including models ranging from 7B to 65B / 70B parameters, and report the complete results in Table 3.9.

3.4 OUR RESULTS VS. OFFICIAL SCORES

To verify the correctness of our implementation, we first compare our evaluation results with the officially reported scores from GPT-4 technical report [45] and Microsoft’s early experiments with GPT-4 [2]. To ensure an apple-to-apple comparison, we align the evaluation settings on each benchmark, as summarized in Table 3.10. This head-to-head comparison demonstrates that our evaluation results are consistent with the official scores, within a

Table 3.8: Benchmark performance with different temperature T and top_p . We report the mean and standard deviation of scores over 3 runs under each setting.

Benchmark	Setting	top_p	code-davinci-002		text-davinci-003		gpt-3.5-turbo-0301	
			$T = 0.5$	$T = 1.0$	$T = 0.5$	$T = 1.0$	$T = 0.5$	$T = 1.0$
MMLU	5-shot	0.2	68.3 ± 0.1	68.3 ± 0.1	65.4 ± 0.1	65.5 ± 0.1	68.4 ± 0.1	68.4 ± 0.0
		0.7	66.9 ± 0.6	65.7 ± 0.5	65.3 ± 0.2	65.4 ± 0.2	68.2 ± 0.1	68.4 ± 0.2
		1.0	65.8 ± 0.0	59.8 ± 0.4	65.2 ± 0.2	65.1 ± 0.3	68.2 ± 0.1	67.9 ± 0.1
GSM8K	8-shot CoT	0.2	60.0 ± 0.7	60.4 ± 0.7	59.6 ± 0.4	59.7 ± 0.5	78.8 ± 0.3	78.6 ± 0.2
		0.7	58.9 ± 1.0	57.3 ± 0.4	59.7 ± 0.5	60.6 ± 0.7	78.9 ± 0.1	78.6 ± 1.1
		1.0	57.7 ± 0.3	31.2 ± 1.5	59.9 ± 1.8	57.2 ± 0.3	78.9 ± 0.1	77.5 ± 0.8
HumanEval	0-shot, pass@1	0.2	29.2 ± 0.0	15.8 ± 0.4	58.5 ± 0.3	55.1 ± 0.4	61.4 ± 0.2	61.3 ± 0.1
		0.7	29.5 ± 0.1	15.6 ± 0.2	58.7 ± 0.2	54.9 ± 0.1	58.0 ± 0.1	57.6 ± 0.2
		1.0	29.4 ± 0.6	15.6 ± 0.4	58.6 ± 0.2	55.3 ± 0.1	57.3 ± 0.1	50.8 ± 0.2
	0-shot, pass@100	0.2	89.4 ± 0.3	88.6 ± 1.8	85.6 ± 1.3	91.5 ± 1.0	62.8 ± 0.0	62.8 ± 0.0
		0.7	88.8 ± 1.4	89.6 ± 1.6	85.1 ± 2.3	91.1 ± 0.1	73.8 ± 0.6	74.4 ± 0.6
		1.0	88.8 ± 0.9	86.8 ± 1.8	87.4 ± 1.8	92.7 ± 1.2	75.2 ± 0.3	79.1 ± 1.0

margin of slight deviation. Since the official prompts and in-context examples for evaluation are not publicly available, the slight deviation is totally reasonable. We also notice that the performance gain with in-context examples beyond 1-shot is pretty marginal, which aligns with our primary evaluation setting in Table 3.1.

We also compare our evaluation results with the official scores reported in LLaMA [17] and Llama 2 [18]. Similarly, in Table 3.11, we report the benchmarks whose official evaluation settings match our settings, and compare our results with the official scores. We observe that on some benchmarks, such as BBH, our results are higher than the official scores; while on some other benchmarks, such as TriviaQA and MATH, our results are lower than the official scores. This phenomenon is consistent with our conclusion that LLaMA and Llama 2 are pretty prompt-sensitive (refer to Table 3.6). To be more specific, take MATH as an example, since we use the exact same setting and prompt as we evaluate OpenAI models on this benchmark, and our evaluation result of GPT-4 matches the official scores (Table 3.10), we argue that the prompt sensitivity of LLaMA / Llama 2 models explains the performance gap of our evaluation and their official scores.

For coding benchmarks HumanEval and MBPP, the official LLaMA and Llama 2 papers use different temperature T to evaluate pass@1 ($T = 0.1$) and pass@100 ($T = 0.8$). In contrast, we follow OpenAI’s setting on coding evaluation [13] and uniformly use $T = 0.8$ for all our evaluations on coding benchmarks. This explains the performance difference of our results and the official scores of LLaMA and Llama 2 on HumanEval and MBPP.

Table 3.9: Complete evaluation results of LLaMA and Llama 2 family models.

Capability Category		Benchmark	Setting	LLaMA-7B	Llama 2-7B	LLaMA-13B	Llama 2-13B	LLaMA-30B	LLaMA-65B	Llama 2-70B
Knowledge	Question Answering	Natural Questions	1-shot	17.6	19.8	20.8	27.6	24.0	27.7	27.0
		WebQuestions	1-shot	37.0	38.3	37.6	42.8	39.0	42.2	38.2
		TriviaQA	1-shot	52.0	61.1	66.6	70.0	73.5	73.4	74.0
	Multi-subject Test	MMLU	5-shot	25.1	41.0	38.5	49.5	51.0	60.1	67.8
		AGIEval-EN	few-shot	19.1	25.7	27.0	35.7	34.7	38.0	44.0
		ARC-e	1-shot	30.0	62.3	67.6	76.4	82.4	87.2	93.4
		ARC-c	1-shot	26.7	48.6	49.1	55.7	60.8	71.8	79.6
Reasoning	Commonsense Reasoning	LAMBADA	1-shot	19.0	38.0	47.0	56.4	32.5	30.9	30.4
		HellaSwag	1-shot	24.6	25.4	28.9	37.2	31.3	47.8	68.4
		WinoGrande	1-shot	50.4	50.2	48.1	52.1	51.3	54.6	69.8
	Comprehensive Reasoning	BBH	3-shot CoT	33.7	38.4	39.1	46.2	49.6	58.2	65.0
Comprehension	Reading Comprehension	RACE-m	1-shot	26.7	45.8	52.4	57.9	65.3	77.0	87.6
		RACE-h	1-shot	29.1	39.5	48.5	55.1	64.1	73.0	85.1
		DROP	3-shot, F1	9.6	7.7	8.7	9.3	9.8	10.0	12.1
Math	Mathematical Reasoning	GSM8K	8-shot CoT	13.9	17.2	18.4	28.6	35.1	53.6	56.4
		MATH	4-shot CoT	0.4	0.1	0.4	0.5	0.5	2.6	3.7
Coding	Coding Problems	HumanEval	0-shot, pass@1	7.0	14.6	9.7	15.8	7.2	10.7	12.7
		MBPP	3-shot, pass@1	23.7	39.2	29.5	46.0	38.5	44.8	58.0
Multilingual	Multi-subject Test	AGIEval-ZH	few-shot	22.3	23.4	23.5	29.7	28.4	31.7	37.9
		C-Eval	5-shot	11.5	10.3	14.8	28.9	10.1	10.7	38.0
	Mathematical Reasoning	MGSM	8-shot CoT	2.7	2.3	2.8	4.1	3.1	3.6	4.0
	Question Answering	TyDi QA	1-shot, F1	2.4	3.6	3.2	4.5	3.8	12.1	18.8
Safety	Truthfulness	TruthfulQA	1-shot	37.6	31.0	29.5	38.0	44.5	51.0	59.4
	Toxicity	RealToxicityPrompts ↓	0-shot	14.5	14.8	14.9	14.8	14.7	14.8	15.0

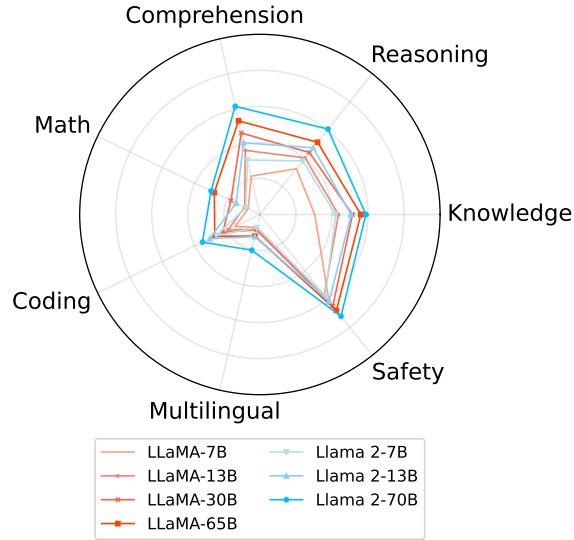


Figure 3.2: Radar charts to visualize the capabilities of LLaMA and Llama 2 family models.

Table 3.10: Comparison of our evaluation results and GPT-4 officially reported scores. The official score of MATH is obtained from [2], which is marked with *.

Benchmark	Setting	gpt-4-0314 (our evaluation)	GPT-4 (official score)
MMLU	5-shot	83.7	86.4
ARC-c	25-shot	96.3	95.6
	1-shot	94.9	–
HellaSwag	10-shot	92.5	95.3
	1-shot	92.4	–
WinoGrande	5-shot	89.3	87.5
	1-shot	86.7	–
DROP	3-shot, F1	78.9	80.9
GSM8K	5-shot CoT	91.6	92.0
	8-shot CoT	92.1	–
MATH	4-shot CoT	38.6	42.5*
HumanEval	0-shot, pass@1	66.3	67.0

Table 3.11: Comparison of our results and the official scores reported in LLaMA and Llama 2 papers.

Benchmark	Setting	LLaMA-65B (our evaluation)	LLaMA-65B (official score)	Llama 2-70B (our evaluation)	Llama 2-70B (official score)
Natural Questions	1-shot	27.7	31.0	27.0	33.0
TriviaQA	1-shot	73.4	84.5	74.0	85.0
MMLU	5-shot	60.1	63.4	67.8	68.9
BBH	3-shot CoT	58.2	43.5	65.0	51.2
GSM8K	8-shot CoT	53.6	50.9	56.4	56.8
MATH	4-shot CoT	2.6	10.6	3.7	13.5
HumanEval	0-shot, pass@1	10.7 ($T = 0.8$)	23.7 ($T = 0.1$)	12.7 ($T = 0.8$)	29.9 ($T = 0.1$)
MBPP	3-shot, pass@1	44.8 ($T = 0.8$)	37.7 ($T = 0.1$)	58.0 ($T = 0.8$)	45.0 ($T = 0.1$)

CHAPTER 4: FACTUALITY BACKGROUND

ChatGPT/GPT-4 [45, 46] has gained substantial recognition for its practical applications, providing useful and informative responses to a wide range of queries. Recent studies have conducted comprehensive technical evaluations of ChatGPT on numerous NLP tasks, demonstrating that ChatGPT outperforms other models across various tasks [2, 47, 48, 49].

However, in spite of the impressive capabilities exhibited by ChatGPT, researchers have highlighted some challenges of ChatGPT, such as its inability to perform reliable reasoning [47], translate low-resource languages effectively [48], solve complex mathematical problems [50], and provide accurate information [47]. While these shortcomings are documented, the specific limitations of ChatGPT that contribute to these challenges are not entirely clear in the existing literature. Taking question answering as a representative example: Is the model’s failure due to its inability to reason or a lack of knowledge to answer the question? Is the issue a result of insufficient knowledge, or does the model struggle to recall the internal knowledge with the question? Is the difficulty in recalling knowledge the root cause, or does the model have trouble understanding the question’s context or intent?

In this study, we delve into an in-depth exploration of the weakness of ChatGPT in the context of complex open-domain question answering, as this task aligns closely with users’ everyday search demands and requires extensive knowledge as well as robust understanding and reasoning capabilities. Our goal is to identify common failure modes of ChatGPT in providing truthful answers, pinpoint the specific abilities in which ChatGPT is deficient that contribute to these failures, and consider potential strategies for improvement.

To this end, we first employ a thematic analysis to analyze instances of ChatGPT’s failures and categorize them into four primary error types: *comprehension error*, *factuality error*, *specificity error*, and *inference error*. We then pinpoint the *factuality* deficiency as the primary failure and identify *knowledge memorization* and *knowledge recall* as critical abilities for answering questions with *factuality*. Furthermore, we propose several potential strategies to help mitigate these deficiencies. Our results indicate that ChatGPT’s *factuality* can be enhanced by supplying granular external knowledge and cues for knowledge recall. Our findings provide practical insights for developing more reliable question answering systems.

CHAPTER 5: TOWARDS FACTUALITY

5.1 MODELS AND DATASETS

We focus on complex open-domain question answering, using two widely-used benchmark datasets: HotpotQA [51] and BoolQ [52], both of which use Wikipedia as their knowledge source. We selected 200 questions from HotpotQA for analyzing the errors made by ChatGPTs. For factuality evaluation, we sampled an additional 500 questions from HotpotQA and 1000 questions from BoolQ. We evaluated the performances of both GPT-3.5 and GPT-4 using these datasets. To generate responses from GPT-3.5 and GPT-4, we utilized the public OpenAI API. We use the model gpt-3.5-0301 and gpt-4-0314.

5.2 CHATGPT’S FAILURES

5.2.1 Thematic Analysis

We examined the model’s responses to 200 HotpotQA samples using thematic analysis [53], a method for identifying patterns or “themes” within data. The process starts by extracting preliminary “codes” from data, which are later assembled into broader themes. For the purpose of ensuring a rigorous and comprehensive thematic analysis, a two-annotator approach was employed. We asked two independent annotators, both proficient in the subject and experienced in qualitative analysis to independently review the data set. Initially, they worked separately to identify the codes, analyze, and report patterns (themes) within the data. Upon completion of their independent analyses, the annotators convened in a collaborative session to compare, discuss, and reconcile any discrepancies in their identified themes. After the inter-annotator discussion, we grouped them several themes and validated these themes with an extra 40 questions. We confirm a complete representation of errors can be achieved through these themes.

To quantify the degree of consistency between the annotators, we employed Cohen’s Kappa κ [54]. This statistical measure is used to evaluate the reliability of agreement between two raters beyond what is expected by chance.

In our study, the computed inter-coder agreement was 0.8394, which suggests a high degree of consensus between the annotators. We finalize the four identified themes as: problem comprehension and intent, factual correctness, specificity level, and reasoning, referred to as *comprehension error*, *factuality error*, *specificity error*, and *inference error*.

Table 5.1: Examples of four types of errors (text in brackets represents the groundtruth, text in blue indicates the entities compared and their information, text in red indicates where the model (GPT-3.5) makes mistakes).

Type	Question and GroundTruth	Responses
Comprehension	What is a rare breed of dog that was derived as a variant of Rat Terrier, <i>Shiloh Shepherd dog</i> or <i>American Hairless Terrier</i> ? (<i>American Hairless Terrier</i>)	One rare breed of dog that was derived as a variant of Rat Terrier , Shiloh Shepherd dog or American Hairless Terrier is the Teddy Roosevelt Terrier.
Factuality	Are the <i>New Orleans Outfall Canals</i> the same length as the <i>Augusta Canal</i> ? (<i>yes</i>)	No... the <i>Orleans Canal</i> is approximately 22 miles (35 kilometers) long ... The <i>Augusta Canal</i> ... is approximately 7 miles (11.3 kilometers) long
Specificity	1. What profession do <i>K'L'ġ'gb'L'ġ Abe</i> and <i>Agatha Christie</i> share? (<i>playwright</i>) 2. What genre do <i>Superheaven</i> and <i>Oceansize</i> belong to? (<i>rock</i>)	1. Author , ... 2. Superheaven and Oceansize are not of the same genre . Superheaven ... play a style of alternative rock ... Oceansize ... was a British progressive rock band ...
Inference	1. Which band has more members, <i>Muse</i> or <i>The Raconteurs</i> ? (<i>The Raconteurs</i>) 2. Which is currently more valuable, <i>Temagami-Lorrain Mine</i> or <i>Meadowbank Gold Mine</i> ? (<i>Meadowbank Gold Mine</i>)	1. Muse has more members than The Raconteurs . <i>Muse</i> is a British rock band with three members ... The Raconteurs ... <i>Raconteurs</i> has four members . 2. I cannot provide the current valuation ... <i>Meadowbank Gold Mine</i> ... was producing gold at a rate of approximately 220,000 ounces per year . On the other hand, <i>Temagami-Lorrain Mine</i> is a historic iron mine located in Ontario, Canada that has been inactive for many years .

Comprehension errors refer to the failure in comprehending the question context and intention. In our experiment, the model demonstrates proficiency in comprehending the question, but it exhibits shortcomings when faced with questions containing grammar mistakes or ambiguity. For instance, the question shown in the Comprehension row of Table 5.1 presents a challenge to the model due to an incorrect interrogative pronoun “what”, which should be “which”. Consequently, the model fails to recognize that the question is seeking a selection between the two items marked in blue and instead misinterprets it as a choice among all last three items.

A *factuality error* occurs when a model lacks the necessary supporting facts to produce an accurate answer [55, 56]. This may be due to the model’s lacking knowledge of a particular

entity, attribute, or event. The example in the Factuality row of Table 5.1 shows a mistake when the model has the incorrect knowledge about the length of two canals. While this type of error is straightforward, it accounts for a majority of errors in the model.

Specificity errors occur when the model fails to answer a problem at the appropriate level of specificity [57, 58]. This can manifest as the model providing an answer that is too general or too specific. For instance, in the specificity row of Table 5.1, for the first question, the correct answer is playwright, but if the model answers with author, it is not specific enough. Similarly, in the second question, the ground truth is rock, but the model predicts [Superheaven](#) as alternative rock and [Oceansize](#) as progressive rock, and claims that they do not share the same genre. It is also classified as specificity error.

An *inference error* occurs when a model possesses the necessary knowledge to answer a question, but fails to reason with the facts effectively to arrive at the correct answer [59, 60]. For instance, for the first question in the inference row of Table 5.1, the model may know that [Muse](#) has three members and [The Raconteurs](#) has four members, but still incorrectly claims that [Muse](#) has more members. Additionally, the model may fail to make predictions based on commonsense. In the second question, although holding the knowledge that the [Meadowbank Gold Mine](#) is still producing gold and the [Temagami-Lorrain Mine](#) has been inactive for years, the model still fails to deduce that the former one is currently more valuable due to its ongoing production.

5.2.2 Results

Table 5.2: Number of failures under different settings.

	#Correct	#Wrong	Comprehension	Factuality	Specificity	Inference
GPT-3.5	130	70	11	46	6	7
GPT-3.5+evi	186	14	1	6	2	5
GPT-4	149	51	6	37	3	5

We counted the frequency of errors across the four categories. We used GPT-3.5 as the baseline model by feeding it plain questions. To study the effect of providing external evidence and to investigate GPT-4’s improvement, we also explored providing questions with gold evidence (accurate piece of information that provides a definitive answer to a question) to the GPT-3.5 model (GPT-3.5+evi) and plain questions to the GPT-4 model. The results are summarized in Table 5.2.

Based on our experiments, we made the following observations: 1) Nearly half of the failures are due to *factuality error*, followed by *inference error*, *comprehension error* and

Table 5.3: Examples of questions and responses on factuality related abilities (The second column demonstrates the erroneous question and response, and third column illustrates our prompt to test the two abilities. Text in blue indicates the model’s responses, and text in red is where the model make a mistake).

Abilities	Erroneous Responses	Test Conversation
Memorization	-> Are the New Orleans Outfall Canals the same length as the Augusta Canal?: -> No...the Orleans Canal is approximately 22 miles (35 kilometers) ...The Augusta Canal...is approximately 7 miles (11.3 kilometers) long	-> User (asking essential knowledge#1): What’s the length of New Orleans Outfall Canals? -> Model: The total length of the New Orleans Outfall Canals is approximately 22.5 miles (36.2 kilometers). -> User (asking essential knowledge#2): What’s the length of Augusta Canal? -> Model: The Augusta Canal is approximately 7 miles (11.3 kilometers) long.
Recall	-> Chang Ucchin was born in korea during a time that ended with the conclusion of what? -> I’m sorry, but I need more information to answer your question accurately.	-> User (asking essential knowledge#1): When was Chang Ucchin born in korea? -> Model: Chang Ucchin ... was born on March 26, 1891... -> User (asking original question): Chang Ucchin was born in korea during a time that ended with the conclusion of what? -> Model: Chang Ucchin was born in Korea during a time when Korea was under Japanese rule, ...ended with the conclusion of World War II in 1945.

specificity error. 2) Providing evidence not only addresses *factuality* but also significantly mitigates *comprehension* and *specificity errors*. 3) The GPT-4 model demonstrates some improvements compared to GPT-3.5, particularly in addressing *comprehension* and *specificity errors*. However, *factuality* is only marginally improved. Our finding underlines *factuality* as the primary concern in open-domain QA for its dominance among errors, its impact on other error types, and GPT-4’s inadequate improvement in this area.

5.3 ABILITIES BEHIND FACTUALITY

As we pinpoint in Section 5.2.2 that *factuality* is the most critical concern in open-domain QA, we focus on investigating factuality. Drawing from cognitive science research on human

memory organization in QA [61], we treat *factuality error* as a cognitive failure in retrieving relevant knowledge from LLMs’ memory for a question.

We define *essential knowledge* p with respect to a question q as *the knowledge indispensable for answering question q* and identify two key abilities for the knowledge retrieval process:

Knowledge Memorization There exist an appropriate prompt s which, when fed into the model, will result in the essential knowledge p .

Knowledge Recall Given the question q as the prompt, the model is able to output the memorized essential knowledge p .

Focusing on these two abilities, we conducted experiments with questions the model couldn’t answer due to lack of knowledge. To test knowledge memorization, we rephrased essential knowledge as a question. E.g., in Table 5.3, we evaluated the model’s memorization by asking about canal lengths. To further evaluate knowledge recall, we re-asked the original question in the same conversation. If the model answered accurately in this setting but not when only asked the original question, it indicated a recall issue. In the recall row of Table 5.3, the model correctly answers the query about Chang’s birth event after a preceding related question, but fails without this context.

We find that 6 out of 46 errors stem from the recall issues, while the rest come from memorization. Our novel insight that distinguishes knowledge memorization and recall offers a new perspective for addressing knowledge-related problems.

5.4 IMPROVING FACTUALITY FOR QA

5.4.1 Settings

In our previous experiment, we demonstrated that a factuality error can stem from knowledge memorization or recall. To investigate how to mitigate it, we conducted experiments using the HotpotQA and BoolQ datasets. We used the plain question configuration with the GPT-3.5 model as our baseline. Since the GPT-3.5 and GPT-4 share similar underlying architectures and training processes, we experimented only on the former to draw observations. The prompts of our experiments are shown in Appendix ???. For evaluation, we employed partial match [62], which examines whether the ground truth is a substring of the prediction.

5.4.2 Knowledge Memorization

The limitations of models in memorizing facts has steered research towards retrieval-augmented language models, such as those augmented with external corpora [63, 64, 65] or search engines [66, 67]. However, noise and non-essential information can compromise retrieval, e.g., notable methods retrieve the whole web pages from Bing search as knowledge [67]. Hence, we refer to the retrieved knowledge as context information and define *granularity* as the length ratio of the context information to the essential knowledge. We investigate the role of *granularity* in performance across four different granularity settings.

- **Sentence level.** Directly provide external knowledge at the sentence level.
- **Passage level.** We offer gold evidence sentences along with other sentences related to the entities.
- **Section level.** We supply the Wikipedia section containing the gold evidence sentences.

5.4.3 Knowledge Recall

To mitigate knowledge recall issues, we considered the knowledge recall process as retrieving the values (essential knowledge) in LLMs’ memory with the given keys (the plain question by default), and tested whether supplying entity-related keys aids the knowledge recall process. Based on the keys provided, we proposed the following settings:

- **Complete entity name.** We give the model complete Wikipedia names of the core entities in the question, e.g., for the film “Samson and Deliah”, we provide “Samson and Deliah (1984 film)”.
- **Definition sentences.** We provide the entity’s initial Wikipedia sentences as the definition or background in addition to the entity names, ensuring no essential knowledge is present.
- **Random relevant sentences.** We also provide other random sentences from the entity’s Wikipedia page along with entity names, again avoiding essential knowledge.

5.4.4 Observations and Implications

Finding I. Finer granularity of external knowledge yields better results. Table 5.4 (top) shows that external knowledge incorporation boosts performance, and effectiveness is influenced by knowledge granularity. Including essential knowledge with other

Table 5.4: Factuality experiments on HotpotQA and BoolQ.

	HotpotQA	BoolQ
Plain Question	0.58	0.71
External - Sentence	0.73	0.869
External - Passage	0.718	-
External - Section	0.603	0.77
Recall - Complete Entity Name	0.597	0.755
Recall - Wiki Background	0.646	0.789
Recall - Random Sentences	0.55	-

sentences affects (although only slightly) results, while using the whole Wikipedia section with evidence greatly reduces the performance gain. This suggests that performance decreases with coarser input knowledge granularity.

Finding II. Providing relevant keys aids in recalling essential knowledge. Table 5.4 (bottom) indicates that supplying the model with keys can enhance accuracy. Supplying the complete entity names improves the performance, while providing entity background or definition sentences further aids knowledge recall, even without the essential knowledge. However, random sentences from the entities’ Wikipedia page do not improve the performance, but instead decrease it.

Based on these findings, we explore strategies to boost the factuality in question answering from a LLM research perspective.

- **Provide external context with fine granularity as evidence to help memorize essential knowledge.** Although LLMs possess a vast amount of implicit knowledge, it is evident that there is still a significant amount of knowledge that is difficult to cover during training or challenging to recall during inference. Therefore, building an IR system to retrieve knowledge with finer granularity based on the question would be helpful according to our findings. Some attempts in this direction have been observed in systems such as New Bing, Bard, and ChatGPT plugins.
- **Provide descriptions of entities as keys to help recall essential knowledge.** In our analysis, we observe that sometimes ChatGPT indeed memorize the essential knowledge to solve the question, but cannot recall the knowledge with the question. A relevant finding is highlighted in the study by [68], where it is observed that while language models can memorize a substantial number of email addresses, they struggle to associate specific email addresses with corresponding individual names. Improving the recall capabilities could be an intriguing direction for developing more powerful language models.

CHAPTER 6: CONCLUSIONS

In the culmination of this thesis, we have systematically dissected the landscape of leading large language models (LLMs) and embarked on a meticulous investigation of their capabilities, particularly focusing on their application as question-answering systems and their adherence to factuality. Through the lens of GPT-Fathom, an innovative evaluation suite, we scrutinized over ten leading LLMs across more than twenty benchmarks, unearthing valuable insights into the evolution and current standings of these models, especially highlighting the trajectory from OpenAI’s GPT-3 to GPT-4. This comprehensive evaluation not only demystifies the performance gap between OpenAI and non-OpenAI models but also illuminates potential pathways for future enhancements in LLM capabilities through strategic data inclusion and training methodologies.

Furthermore, our in-depth exploration into ChatGPT/GPT-4’s performance on complex open-domain question answering tasks reveals critical insights into its operational weaknesses. By categorizing errors into comprehension, factuality, specificity, and inference errors, we identified factuality as a pivotal area where ChatGPT often stumbles. Our findings suggest that augmenting ChatGPT with granular external knowledge and enhancing knowledge recall mechanisms can significantly improve its factuality, thereby making it a more reliable question-answering system.

As we stand at the crossroads of LLM development, this thesis underscores the importance of a systematic, transparent, and reproducible evaluation framework like GPT-Fathom, which serves as a benchmark for current and future LLMs. It also highlights the critical need for continued research into understanding and mitigating the limitations of these models, particularly in areas that directly impact their usability and reliability in real-world applications.

The synthesis of these studies presents a clear roadmap for future research and development in the field of LLMs. By addressing the highlighted challenges and implementing the proposed strategies, we can pave the way for more accurate, reliable, and comprehensive LLMs. These advancements will not only enhance the capabilities of LLMs in question-answering contexts but also contribute to the broader goal of achieving Artificial General Intelligence (AGI) with a strong foundation in factuality and reasoning. As we move forward, it is imperative that the AI research community continues to prioritize transparency, reproducibility, and ethical considerations in the development and evaluation of LLMs, ensuring that these powerful tools serve the best interests of humanity.

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